SOUMYADEEP ROY

Postdoctoral Scholar, Stanford University
https://roysoumya.github.io
soroy@stanford.edu

BIO

Postdoctoral Researcher specializing in natural language processing and foundation models for healthcare applications. Expertise in developing and evaluating medical AI systems with focus on reliability, efficiency and interpretability. Published 15+ peer-reviewed papers in top-tier venues (SIGIR, ACL, IJCAI, EMNLP, ECAI, CIKM) on medical language understanding, efficient model training and clinical NLP applications.

RESEARCH INTERESTS

Foundation Models for Medicine · Medical NLP · Model Evaluation & Robustness · Efficient Pretraining · Clinical Decision Support · Text Summarization · Generative AI.

EDUCATION

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR, Kharagpur, India
PhD, Computer Science and Engineering — 07/2025

- Thesis: Domain Adaptation for Medical Language Understanding.

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR, Kharagpur, India
MS, Computer Science and Engineering, CGPA: 9.21 — 11/2019

- Thesis: Computational Approaches for Online Reputation Monitoring [Modeling, Analysis and Recommendation].

KALYANI GOVERNMENT ENGINEERING COLLEGE, KALYANI, West Bengal, India
B.Tech, Computer Science and Engineering, CGPA: 8.61 — 06/2017

- Thesis: Understanding Email Interactivity and Predicting User Response to Email.

EXPERIENCE

STANFORD UNIVERSITY, United States
Postdoctoral Scholar, Division of Computational Medicine, Dept. of Medicine — 09/2025 - Present

- Faculty Advisor: Prof. Tina Hernandez-Boussard.
- Developing auditable medical AI systems that use clinical guidelines as evidence source.
- Improving efficiency and transparency of reasoning LLMs.

WIPRO GE HEALTHCARE PVT. LTD, Bangalore, India
PhD AI Intern, HealthCare Technology & Innovation Center — 11/2024 - 05/2025

- Developed generative AI-based dashboard for oncology patient data using Chroma vector database and LangChain, enabling clinicians to efficiently navigate complex patient histories through semantic search.
- Built deep learning-based representation learning system for sensor log data from medical imaging equipment to enable anomaly detection and predictive maintenance. Resulted in one patent filing.
- Collaborated with multi-disciplinary research teams to develop Proof-of-Concept (PoC) models that are ready for engineering teams to make them production-ready.

LEIBNIZ UNIVERSITY HANNOVER, Hannover, Germany
Research Associate, L3S Research Center 01/2021 - 06/2023

- Led multi-institutional research collaboration with Hannover Medical School on Parkinson's Disease patient subtyping using Michael J. Fox Foundation LRRK2 dataset (1,500+ patients with clinical data).
- Developed interpretable decision tree-based patient stratification method validated by neurologists, published in *Frontiers in Artificial Intelligence* journal.
- Developed the pointwise mutual information-based token masking technique for DNA transformer models such as DNABert and LOGO. Achieved 10x-speedup for few-shot gene sequence classification task.

ADOBE SYSTEMS INDIA PVT. LTD, Bangalore, India
Research Intern, Big Data Experience Lab 05/2018 – 06/2018

- Built deep learning-based classification model for brand personality detection from website content of Fortune 500 companies.
- Designed and executed large-scale data annotation project on Amazon Mechanical Turk with quality control mechanisms.

KEY RESEARCH CONTRIBUTIONS

Medical LLM Evaluation & Error Analysis
[SIGIR 2024] Conducted systematic evaluation of GPT-4 on USMLE medical licensing exam questions, creating comprehensive error taxonomy identifying 8 distinct failure modes. Released publicly available dataset with 250+ annotated errors doi code.
[ACL 2025 Findings] Demonstrated that high out-of-vocabulary concentration in medical texts leads to significant performance degradation in summarization tasks, with vocabulary adaptation providing effective mitigation doi code.
[WSDM 2025] Co-organized tutorial "Building Trustworthy AI Models for Medicine: From Theory to Applications" covering evaluation protocols, robustness assessment, and deployment considerations doi tutorial website.

Efficient Model Training & Domain Adaptation
[ECAI 2024] Designed adaptive, curriculum masking strategy for DNA transformer models to achieve better performance on genome understanding evaluation dataset, for both full dataset and few-shot gene sequence classification doi code.
[ECAI 2023] Developed GeneMask, a masked language modeling masking technique, to jointly mask highly correlated tokens, to reduce model pretraining time, and achieve 10x speedup enabling effective few-shot classification of genomic doi code.
[IJCAI 2024] Proposed vocabulary adaptation method for medical text summarization, reducing fine-tuning time from 450 days to 45 hours through fragment score-based hyperparameter optimization doi code.
[EMNLP 2024 Findings] Developed adaptive BPE tokenization technique that modifies initialization step to better preserve domain-specific terminology during fine-tuning doi code.

Clinical NLP Applications
[ICDH 2023] Developed interpretable clinical trial search system using PubMed citation network to improve retrieval by metapath-based similarity search and improved aspect-based ranking fusion (Best Student Paper Candidate) doi code.
[CIKM 2021] Built knowledge-aware dual-encoder neural network for medical forum question classification, prioritizing medical concept-bearing words extracted using the QuickUMLS tool doi code.

[ECAI 2025] Mentored a PhD student through a multi-institution collaboration, creating annotated dataset for physician intent modeling in doctor-patient dialogues and showing intent filtering improves dialogue summarization performance doi code.
[Frontiers in AI 2025] Published decision tree-based approach for Parkinson's Disease patient subtyping using clinical and genomic data, validated by medical experts doi code.

TECHNICAL SKILLS

Deep Learning: PyTorch, Transformers, Parameter-Efficient Fine-Tuning (LoRA, QLoRA).
NLP & LLMs: Fine-tuning, Prompt Engineering, RAG Systems, Model Evaluation, Efficient Training.
Healthcare AI: Clinical NLP, EHR Analysis, Medical Knowledge Graphs (UMLS, PrimeKG), MIMIC-III/IV.
ML Tools: Weights & Biases, Docker, Git.
Programming: Python, R, SQL, Bash.
Data Collection & Analysis: Potato, Amazon Mechanical Turk, Prolific.

PROFESSIONAL ACTIVITIES

Reviewing

- Reviewed for CORE A* conferences: EMNLP 2022, ACL 2023, EMNLP 2023, AAAI 2023 to 2026, ACL Rolling Review February, May, July 2025, January 2026, FAccT 2026.
- Reviewed for other conferences: COLING 2025, ML4H 2024, EACL 2023, ECML-PKDD 2020 (CORE A).
- Reviewed for journals - Journal of the American Medical Informatics Association, Frontiers in AI, Knowledge and Information Systems, Frontiers in Genetics, IEEE Transactions on Cognitive and Developmental Systems, Elsevier Artificial Intelligence in Medicine.

Tutorials/Invited Talks

- January 2026: Invited talk titled "Building a Career in AI for Industry and Academia - Current and Future trends" organized by the Google Developers Group on Campus Kalyani Government Engineering College - Kalyani, India Link to website, Link to slides.
- December 2025: Invited talk titled "Vocabulary Adaptation Strategies for Finetuning Medical Language Models - Towards Reliable Medical Summarization" delivered at the Breakfast Talk series of Microsoft Research India Bangalore, India Link to slides.
- August 2025: Invited to present ECAI 2023 and 2024 work on "Improving Training Efficiency of DNA Foundation Models" at the 1st Symposium on Intersections: Statistical Genomics and Complex/Chronic Disease Biology, organized by BRICS - National Institute of Biomedical Genomics, Kalyani, West Bengal.
- March 2025: Conference tutorial titled "Building Trustworthy AI Models for Medicine: From Theory to Applications" delivered at the 18th ACM International Conference on Web Search and Data Mining (WSDM 2025) Tutorial website, Link to slides.
- December 2024: Delivered an invited talk on "Foundation Models for Medical Text", for the course "AI Foundation Models in Biomedicine" of Leibniz University Hannover, Germany Link to slides.

Teaching

- **IIT Kharagpur:** Programming and Data Structure Lab (Spring 2018), Department Website Team (Autumn 2018 to Spring 2020), Information Retrieval (Autumn 2020), Computing Lab 1 (Autumn 2023).
- **Leibniz University Hannover, Germany:** Natural Language Processing (Winter 2021), Foundations of Information Retrieval (Summer 2022, 2023).